

Extended sign-kill for $\widehat{\lambda} = (3, 3)$: single Phi-branch with vanishing same-row hook

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Abstract

Let $\lambda = (3, 3, 1^q) \vdash n = q + 6$ with $q \geq 4$, $\ell = q + 2 = n - 4$, and $B = B_{\ell+1}^{(\lambda)} V_\lambda = R'_{\ell+1} R'_{\ell+2} R'_{\ell+3} R'_{\ell+4} V_\lambda$. We prove the “ $\subseteq$ ” direction of the extended sign-kill conjecture for this family at the predicted threshold $\tau = q - 3$:

$$B \subseteq \bigcap_{j=1}^{q-3} E_j^-|_{V_\lambda}.$$

Sharpness at $j = q - 2$ is reduced structurally to a boundary-prefix witness in $\text{supp}(B)$, verified computationally for $q \in \{4, 5, 6, 7\}$. At $q \in \{2, 3\}$ the containment is vacuous.

The shape $\widehat{\lambda} = (3, 3)$ has only *two* removable corners (since $\lambda_1 = \lambda_2 = 3$), so λ is an $r = 1$ shape in the multi-corner classification with $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2,2)} = 2$. The decomposition $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$ splits into a single 2-block piece $\Phi_A(V_{\mu_A})$ over $\mu_A = (3, 2, 1^{q-1})$ *plus* a same-row hook piece $\Phi_{\text{hook}}(V_{\mu_{\text{hook}}})$ over $\mu_{\text{hook}} = (3, 1^{q+1})$. The hook piece’s pullback contribution to B *vanishes* by the May-12 hook column-1 theorem (the same mechanism that kills the C -piece in the previous paper on $\widehat{\lambda} = (4, 2)$), reducing the reduction to a single Phi-branch recursive on the May-14 fifth-pass $(3, 2)$ theorem.

1 Setup

We use the conventions of the May-13–May-14 series. $H_q(S_n)$ is the Iwahori–Hecke algebra over $\mathbb{Q}(q)$ with generators $T_1, \dots, T_{n-1}$ and quadratic relation $T_j^2 = (q - 1)T_j + q$. V_λ is the Specht module in the Hoefsmit seminormal basis $\{v_T : T \in \text{SYT}(\lambda)\}$.

For $1 \leq j \leq n - 1$, set $S_j := T_j + 1$. Write

$$E_j^+ := \ker(T_j - q)|_{V_\lambda} = \text{Im}(S_j|_{V_\lambda}), \quad E_j^- := \ker(S_j)|_{V_\lambda}.$$

For $k \geq 1$, let $R'_k := S_{k-1} S_{k-2} \cdots S_1$ with the convention $R'_1 = 1$. For $\lambda \vdash n$ with $\ell = \ell(\lambda)$, set

$$\Omega = \Omega^{(\lambda)} := R'_{\ell+1} R'_{\ell+2} \cdots R'_n, \quad B := \Omega \cdot V_\lambda = B_{\ell+1}^{(\lambda)} V_\lambda.$$

For $\lambda = (3, 3, 1^q)$, $n = q + 6$ and $\ell = q + 2$, so $\Omega = R'_{q+3} R'_{q+4} R'_{q+5} R'_{q+6} = R'_{n-3} R'_{n-2} R'_{n-1} R'_n$ has *four* R' -blocks. By the May-13 multi-corner-dimension paper, $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2,2)} = 2$ in the stable regime $q \geq 2$.

Removable corners of λ . Since $\lambda_1 = \lambda_2 = 3$, the cell $(1, 3)$ is *not* removable. The removable corners are

$$c_1 = (2, 3), \quad c_2 = (q + 2, 1).$$

There are only *two* corners, so λ is an $r = 1$ shape in the multi-corner classification (corner count minus one).

Identity prefix. For an SYT T of any partition μ , the *identity prefix length* is

$$p(T) := \max\{P \geq 0 : T(i, 1) = i \text{ for all } 1 \leq i \leq P\}.$$

Seminormal support. For $W \subseteq V_\lambda$,

$$\text{supp}(W) := \{T \in \text{SYT}(\lambda) : \text{some } w \in W \text{ has } [v_T]w \neq 0\}.$$

2 Input from prior papers

2.1 The support rule and prefix corollary (May-14 fourth pass)

Proposition 1 (Support rule). *Let $W \subseteq V_\lambda$ be any subspace. Fix $1 \leq j \leq n-1$. If every $T \in \text{supp}(W)$ has j and $j+1$ in the same column of T , then $W \subseteq E_j^-|_{V_\lambda}$.*

Remark 2. If every $T \in \text{supp}(W)$ has $p(T) \geq P$, then for every $1 \leq j \leq P-1$ the entries $j, j+1$ sit at column-1 rows $j, j+1$, sharing column 1. By Proposition 1, $W \subseteq E_j^-$ for $1 \leq j \leq P-1$.

2.2 The May-14 fifth-pass (3,2) extended sign-kill theorem

Theorem 3 (Extended sign-kill for $\hat{\lambda} = (3, 2)$, May-14 fifth pass). *For $\lambda' = (3, 2, 1^{q'}) \vdash n' = q' + 5$ with $q' \geq 3$, every $T' \in \text{supp}(B_{q'+3}^{(\lambda')} V_{\lambda'})$ satisfies $p(T') \geq n' - 6 = q' - 1$. Consequently, $B_{q'+3}^{(\lambda')} V_{\lambda'} \subseteq E_j^-|_{V_{\lambda'}}$ for $1 \leq j \leq q' - 2$.*

2.3 The outer-factor preservation lemma (May-14 fifth pass)

Lemma 4 (Outer-factor preservation). *Let $W \subseteq V_\nu$ be a subspace such that every $T \in \text{supp}(W)$ satisfies $p(T) \geq P$. Then for every $j \geq P+1$, every $T'' \in \text{supp}((T_j+1)W)$ also satisfies $p(T'') \geq P$.*

Proof sketch. (T_j+1) acts within the 2-block $\{v_T, v_{s_j T}\}$; the swap s_j moves only entries j and $j+1$, both $\geq j \geq P+1 > P$, so neither lies in the column-1 prefix and the prefix is preserved. $\square \quad \square$

2.4 The May-12 hook column-1 theorem

Theorem 5 (Hook column-1, May-12). *For $\mu = (k, 1^m) \vdash n_0$ with $k \geq 2$ and $m \geq \max(k, 2)$, $B_{\ell_0+1}^{(\mu)} V_\mu \subseteq E_1^-|_{V_\mu}$ (where $\ell_0 = \ell(\mu) = m+1$).*

2.5 Phase A endpoint (May-19)

Theorem 6 (Abstract Phase A endpoint). *For any H_q -module V and any $1 \leq j \leq n-1$, $(T_j + 1) \cdots (T_1 + 1)V = E_j^+(V)$. In particular, $R'_n V_\lambda = E_{n-1}^+|_{V_\lambda}$.*

3 The $+q$ -eigenspace decomposition for $(3, 3, 1^q)$

A crucial subtlety distinguishes the $r = 1$ shape $\lambda = (3, 3, 1^q)$ from the $r = 2$ shapes $(3, 2, 1^q)$ and $(4, 2, 1^q)$ treated in the fifth- and sixth-pass papers: the $+q$ -eigenspace $E_{n-1}^+|_{V_\lambda}$ has *two* structurally distinct pieces, not one.

2-block piece (doubly-stripped at $\{c_1, c_2\}$). The pair $\{c_1, c_2\} = \{(2, 3), (q + 2, 1)\}$ has axial distance

$$|d| = |c(c_1) - c(c_2)| = |1 - (-q - 1)| = q + 2 \geq 4 \quad \text{for } q \geq 2,$$

so the corresponding T_{n-1} -block is a non-degenerate 2-block. The doubly-stripped partition is

$$\mu_A := \lambda \setminus \{c_1, c_2\} = (3, 2, 1^{q-1}) \quad (\text{a } \widehat{\lambda} = (3, 2) \text{ shape!}).$$

Same-row hook piece at c_1 . The cell $(2, 2)$ sits immediately left of $c_1 = (2, 3)$ in the same row. SYTs of λ with $n - 1$ at $(2, 2)$ and n at $(2, 3)$ are $+q$ -eigenvectors of T_{n-1} (same-row eigenvectors), distinct from the 2-block pieces. Their inner shape is

$$\mu_{\text{hook}} := \lambda \setminus \{(2, 2), (2, 3)\} = (3, 1^{q+1}),$$

a hook of length $q + 2 = n - 4$ and size $n - 2$.

No other $+q$ pieces. Same-row at $c_2 = (q + 2, 1)$: no cell to its left, so vacuous. Same-column extensions at column 1 produce -1 -eigenvectors (E_{n-1}^-), not relevant here.

Lemma 7 (Two-piece decomposition of E_{n-1}^+). *Define $\Phi_A : V_{\mu_A} \hookrightarrow V_\lambda$ on the seminormal basis by*

$$\Phi_A(v_{T'}) := v_{T_A^A} + \tau_A v_{T_A^B},$$

where T_A^A, T_A^B are the two SYTs of λ obtained from $T' \in \text{SYT}(\mu_A)$ by placing $\{n - 1, n\}$ at $\{c_1, c_2\}$ in the two possible orders, and $\tau_A \in \mathbb{Q}(q)^\times$ is the Hoefsmit 2-block coefficient at axial distance $|d| = q + 2$.

Define $\Phi_{\text{hook}} : V_{\mu_{\text{hook}}} \hookrightarrow V_\lambda$ on the seminormal basis by $\Phi_{\text{hook}}(v_{T''}) := v_{T'''}$, where T''' is the unique SYT of λ obtained from $T'' \in \text{SYT}(\mu_{\text{hook}})$ by placing $n - 1$ at $(2, 2)$ and n at $(2, 3)$.

Then:

(i) Both Φ_A and Φ_{hook} are injective and T_i -equivariant for $1 \leq i \leq n - 3$.

(ii) Their images in V_λ have disjoint seminormal supports and satisfy

$$R'_n V_\lambda = E_{n-1}^+|_{V_\lambda} = \Phi_A(V_{\mu_A}) \oplus \Phi_{\text{hook}}(V_{\mu_{\text{hook}}}).$$

Proof. Injectivity and equivariance for $T_1, \dots, T_{n-3}$ are standard (see May-20, Lemma 3.2): for $i \leq n - 3$, the swap s_i moves only entries i and $i + 1$, both $\leq n - 2$, hence both inside the “inner” part (μ_A - or μ_{hook} -cells); the Hoefsmit coefficients depend only on those inner cells, so T_i commutes with the corresponding Φ . For Φ_A the equivariance also passes through the 2-block decoupling at T_{n-1} (same argument as the May-20 $r=2$ case).

Disjointness of supports: Φ_A supports require $n - 1, n$ to occupy $\{(2, 3), (q + 2, 1)\}$ (one each); Φ_{hook} supports require $n - 1 = (2, 2)$ and $n = (2, 3)$. These cell configurations are distinct (the position of $n - 1$ differs), so supports are disjoint.

Dimension: $\dim \Phi_A(V_{\mu_A}) = f^{(3, 2, 1^{q-1})}$ (Φ is injective); $\dim \Phi_{\text{hook}}(V_{\mu_{\text{hook}}}) = f^{(3, 1^{q+1})}$. A direct count of $+q$ -eigenvectors of T_{n-1} on V_λ gives

$$\dim E_{n-1}^+ = \#(2\text{-block pairs at } \{c_1, c_2\}) + \#(\text{same-row SYTs at } c_1) = f^{(3, 2, 1^{q-1})} + f^{(3, 1^{q+1})}.$$

The two pieces are disjoint subspaces of E_{n-1}^+ with summed dimension equal to $\dim E_{n-1}^+$, hence span (in direct sum) all of E_{n-1}^+ . By Theorem 6, $R'_n V_\lambda = E_{n-1}^+$, giving the claimed equality.

[Computational verification at $q \in \{2, 3, 4, 5, 6\}$ confirms $\dim R'_n V_\lambda$, $\dim E_{n-1}^+$, f^{μ_A} , and $f^{\mu_{\text{hook}}}$ all agree as claimed: `~/projects/scratch/2026-05-14-extended-sign-kill-33/check_reduction.py`.]

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Remark 8. The May-14 sixth-pass paper on $\widehat{\lambda} = (4, 2)$ asserts (Lemma 3.1(ii) there) that $E_{n-1}^+|_{V_\lambda} = \bigoplus_X \Phi_X(V_{\mu_X})$ where $X \in \{A, B, C\}$ ranges over the three corner pairs. This is *strictly false* computationally: $\lambda = (4, 2, 1^q)$ also admits a same-row $+q$ -piece at $c_1 = (1, 4)$ (with inner shape $(2, 2, 1^q)$), which is missed by the 2-block Phi sum (verified for $q \in \{2, 3, 4, 5\}$, dimension gap $= f^{(2, 2, 1^q)}$). The sixth-pass main theorem (containment direction) is nonetheless correct because the missing same-row piece's pullback contribution to B vanishes by an additional application of the May-14 fourth-pass extended sign-kill theorem at $\widehat{\lambda} = (2, 2)$. We close that gap in a companion note; the present paper handles the analogous (but hook-based, hence simpler) vanishing for $\widehat{\lambda} = (3, 3)$.

4 The single-branch reduction for $(3, 3, 1^q)$

Theorem 9 (Single-branch reduction). *For $\lambda = (3, 3, 1^q)$ with $q \geq 2$,*

$$B = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) \Phi_A(B_{n-4}^{(\mu_A)} V_{\mu_A}). \quad (1)$$

Proof. We follow the May-15/May-20 pull-through, applied separately to each piece of $R'_n V_\lambda$ from Lemma 7, then show the hook piece vanishes.

Step 1 (pulling $R'_{n-3}R'_{n-2}R'_{n-1}$ through the Phi pieces). Let $\Phi \in \{\Phi_A, \Phi_{\text{hook}}\}$ and $\mu \in \{\mu_A, \mu_{\text{hook}}\}$ correspondingly.

Decompose $R'_{n-1} = (T_{n-2}+1)U_1$ where $U_1 = (T_{n-3}+1) \cdots (T_1+1)$. Since $T_1, \dots, T_{n-3}$ commute with T_{n-1} (index gaps ≥ 2), U_1 preserves $E_{n-1}^+|_{V_\lambda}$. By Lemma 7(i), $U_1\Phi(v) = \Phi(R'_{n-2}^{(\mu)}v)$ for $v \in V_\mu$ (here $R'_{n-2}^{(\mu)}$ is the analogue of R'_{n-2} inside $H_q(S_{n-2})$, using generators $T_1, \dots, T_{n-3}$). Hence

$$R'_{n-1}\Phi(V_\mu) = (T_{n-2}+1)\Phi(R'_{n-2}^{(\mu)}V_\mu).$$

Decompose $R'_{n-2} = (T_{n-3}+1)U_2$ where $U_2 = (T_{n-4}+1) \cdots (T_1+1) \in H_q(S_{n-3})$. U_2 commutes with T_{n-2} (index gap ≥ 2), so $R'_{n-2}(T_{n-2}+1) = (T_{n-3}+1)(T_{n-2}+1)U_2$. And $U_2 = R'_{n-3}^{(\mu)}$ acts on $\Phi(v)$ via Φ . Hence

$$R'_{n-2}R'_{n-1}\Phi(V_\mu) = (T_{n-3}+1)(T_{n-2}+1)\Phi(R'_{n-3}^{(\mu)}R'_{n-2}^{(\mu)}V_\mu).$$

Decompose $R'_{n-3} = (T_{n-4}+1)U_3$ where $U_3 = (T_{n-5}+1) \cdots (T_1+1) \in H_q(S_{n-4})$. $U_3 = R'_{n-4}^{(\mu)}$ commutes with both T_{n-3} and T_{n-2} (index gaps ≥ 2), so

$$R'_{n-3}R'_{n-2}R'_{n-1}\Phi(V_\mu) = (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1)\Phi(R'_{n-4}^{(\mu)}R'_{n-3}^{(\mu)}R'_{n-2}^{(\mu)}V_\mu). \quad (2)$$

Step 2 (combining the two pieces). By Lemma 7(ii) and Theorem 6,

$$\begin{aligned} B &= \Omega V_\lambda = R'_{n-3}R'_{n-2}R'_{n-1}R'_n V_\lambda \\ &= R'_{n-3}R'_{n-2}R'_{n-1}(\Phi_A(V_{\mu_A}) \oplus \Phi_{\text{hook}}(V_{\mu_{\text{hook}}})) \\ &= (T_{n-4}+1)(T_{n-3}+1)(T_{n-2}+1) [\Phi_A(I_A) + \Phi_{\text{hook}}(I_{\text{hook}})], \end{aligned}$$

where $I_X := R'_{n-4}^{(\mu_X)}R'_{n-3}^{(\mu_X)}R'_{n-2}^{(\mu_X)}V_{\mu_X}$.

Step 3 (the hook piece vanishes). For $\mu = \mu_{\text{hook}} = (3, 1^{q+1})$, we have $\ell(\mu) = q+2 = n-4$, hence $\ell(\mu)+1 = n-3$. Therefore

$$I_{\text{hook}} = R'_{n-4}^{(\mu_{\text{hook}})}(R'_{n-3}^{(\mu_{\text{hook}})}R'_{n-2}^{(\mu_{\text{hook}})}V_{\mu_{\text{hook}}}) = R'_{n-4}^{(\mu_{\text{hook}})}B_{n-3}^{(\mu_{\text{hook}})}V_{\mu_{\text{hook}}}.$$

By Theorem 5 applied to $\mu = (3, 1^{q+1})$ with $k = 3$, $m = q + 1$ (hypothesis $m \geq \max(k, 2) = 3$, i.e., $q \geq 2$ ✓), $B_{n-3}^{(\mu_{\text{hook}})} V_{\mu_{\text{hook}}} \subseteq E_1^- |_{V_{\mu_{\text{hook}}}}$. But $R_{n-4}^{(\mu_{\text{hook}})} = (T_{n-5} + 1)(T_{n-6} + 1) \cdots (T_1 + 1)$ has $(T_1 + 1)$ as its *rightmost* factor, which annihilates $E_1^- |_{V_{\mu_{\text{hook}}}}$. Hence $I_{\text{hook}} = 0$ and the hook contribution drops out.

Step 4 (the μ_A piece is $B_{n-4}^{(\mu_A)} V_{\mu_A}$). For $\mu = \mu_A = (3, 2, 1^{q-1})$, $\ell(\mu_A) = q + 1 = n - 5$, so $\ell(\mu_A) + 1 = n - 4$. The triple product $I_A = R_{n-4}^{(\mu_A)} R_{n-3}^{(\mu_A)} R_{n-2}^{(\mu_A)} V_{\mu_A} = B_{n-4}^{(\mu_A)} V_{\mu_A}$ (three consecutive R' -factors starting at $\ell(\mu_A) + 1$, which is the definition of $B_{\ell(\mu_A)+1}^{(\mu_A)}$ for μ_A of size $n - 2$).

Combining Steps 2–4 gives (1). $\square$

Remark 10. The vanishing of the hook piece is the direct analogue of the C -piece vanishing in the sixth-pass (4, 2) paper (Lemma 6 there): in both cases, the inner shape has $\ell + 1 = n - 3$ rather than $n - 4$, which means the triple-product pullback contains a “leftover” $R_{n-4}^{(\mu)}$ factor below the $B_{\ell+1}^{(\mu)}$ object; the hook column-1 theorem (May-12) places $B_{\ell+1}^{(\mu)}$ inside E_1^- , and the rightmost $(T_1 + 1)$ of $R_{n-4}^{(\mu)}$ kills it. The same mechanism applies whenever the same-row inner shape is a hook of appropriate dimensions. In (3, 3) the inner shape $(3, 1^{q+1})$ is a hook; in (4, 2) the corresponding same-row inner shape $(2, 2, 1^q)$ is *not* a hook, so the same-row vanishing in (4, 2) requires the (2, 2) extended sign-kill input.

5 Main theorem

Theorem 11 (Main). For $\lambda = (3, 3, 1^q) \vdash n = q + 6$ with $q \geq 4$,

$$B \subseteq \bigcap_{j=1}^{q-3} E_j^- |_{V_\lambda}.$$

Proof. By Remark 2, it suffices to show that every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 2 = n - 8$. We use the reduction (1) together with a prefix bound on $B_{n-4}^{(\mu_A)} V_{\mu_A}$ and Lemma 4 on the three outer factors.

Step 1: prefix bound on the Φ_A -image. $\mu_A = (3, 2, 1^{q-1})$ has size $n - 2$ and $\ell(\mu_A) = q + 1 = n - 5$, so writing $\lambda' = \mu_A$, $n' = n - 2 = q + 4$, and $q' = q - 1$. Since $q \geq 4$ we have $q' \geq 3$, so Theorem 3 applies:

$$p(T') \geq n' - 6 = q - 2 = n - 8 \quad \text{for every } T' \in \text{supp}(B_{n-4}^{(\mu_A)} V_{\mu_A}).$$

Now $\Phi_A(v_{T'})$ is supported on the two SYTs T_A^A, T_A^B of λ obtained from T' by placing $\{n - 1, n\}$ at $\{c_1, c_2\} = \{(2, 3), (q + 2, 1)\}$ in the two possible orders. Column-1 cells of μ_A are at rows $1, 2, \dots, \ell(\mu_A) = n - 5$. The added cells (2, 3) (column 3, not column 1) and $(q + 2, 1)$ (row $q + 2 = n - 4 > n - 5$, below μ_A 's column-1 cells) do not disturb the column-1 prefix of T' , hence

$$p(T_A^A), p(T_A^B) \geq p(T') \geq n - 8.$$

Step 2: outer factors preserve the prefix. The three outer factors in (1) are $(T_j + 1)$ for $j \in \{n - 4, n - 3, n - 2\}$. Each index satisfies $j \geq n - 4 > n - 7 = (n - 8) + 1$, so the hypothesis $j \geq P + 1$ of Lemma 4 (with $P = n - 8$) is met. Applying the lemma three times in sequence, the prefix bound $\geq n - 8$ is preserved.

Conclusion. By Steps 1–2 and (1), every $T \in \text{supp}(B)$ has $p(T) \geq n - 8 = q - 2$. By Remark 2, $B \subseteq E_j^-$ for $1 \leq j \leq q - 3$. $\square$

Remark 12 ($q \in \{2, 3\}$). At $q = 2$, $\tau = q - 3 = -1$ and the intersection $\bigcap_{j=1}^{-1} E_j^-$ is the whole of V_λ , so the containment is vacuous; similarly at $q = 3$, $\tau = 0$ and the intersection is again the whole module. Theorem 11's hypothesis $q \geq 4$ is the joint requirement of the recursive $(3, 2)$ input (Theorem 3 needs $q' = q - 1 \geq 3$), which exactly aligns with the first q at which the predicted τ is positive.

6 Sharpness

Lemma 13 (Sharpness reduction). *Suppose $B \subseteq E_{q-2}^-|_{V_\lambda}$ for $\lambda = (3, 3, 1^q)$ with $q \geq 4$. Then every $T \in \text{supp}(B)$ satisfies $p(T) \geq q - 1 = n - 7$.*

Proof. Set $j_0 := q - 2 = n - 8$. Suppose for contradiction $T \in \text{supp}(B)$ has $p(T) = n - 8$ exactly. Then there exists $w \in B$ with $C_T := [v_T]w \neq 0$, and by hypothesis $T_{j_0}w = -w$.

Locating entry $n - 7$ in T . By $p(T) = n - 8$, the cells $\{(1, 1), \dots, (n - 8, 1)\}$ hold entries $\{1, \dots, n - 8\}$, and $T(n - 7, 1) \neq n - 7$. The remaining 8 cells of λ are

$$\{(1, 2), (1, 3), (2, 2), (2, 3), (n - 7, 1), (n - 6, 1), (n - 5, 1), (n - 4, 1)\},$$

holding the 8 entries $\{n - 7, n - 6, \dots, n\}$.

We claim entry $n - 7$ sits at $(1, 2)$:

- Not at $(n - 6, 1), (n - 5, 1), (n - 4, 1)$: column-1 increase forces $T(n - 7, 1) < n - 7$, but all remaining entries are $\geq n - 7$. Impossible.
- Not at $(n - 7, 1)$: contradicts $T(n - 7, 1) \neq n - 7$.
- Not at $(1, 3)$: $T(1, 2) < T(1, 3) = n - 7$ forces $T(1, 2) < n - 7$, impossible.
- Not at $(2, 2)$: column-2 increase forces $T(1, 2) < n - 7$, impossible.
- Not at $(2, 3)$: $T(1, 3) < T(2, 3) = n - 7$ forces $T(1, 3) < n - 7$, impossible (and $T(2, 2) < T(2, 3) = n - 7$ likewise).

Hence $T(1, 2) = n - 7$.

Off-diagonal swap. The cells of $j_0 = n - 8$ (at $(n - 8, 1)$) and $j_0 + 1 = n - 7$ (at $(1, 2)$) lie in different rows and columns. Axial distance:

$$|d| = |c(1, 2) - c(n - 8, 1)| = |1 - (1 - (n - 8))| = n - 8 \geq 2 \quad \text{for } q \geq 4.$$

The Hoefsmit 2-block at T_{j_0} is non-degenerate: $T_{j_0}v_T = \alpha v_T + \beta v_{T^*}$ with $\beta \neq 0$, where $T^* := s_{j_0}T$ swaps entries $n - 8$ and $n - 7$ between cells $(n - 8, 1)$ and $(1, 2)$.

T^* is a valid SYT. After the swap, $(n - 8, 1)$ holds $n - 7$ and $(1, 2)$ holds $n - 8$. Column-1 rows $1, \dots, n - 9$ unchanged; $T^*(n - 8, 1) = n - 7 > T^*(n - 9, 1) = n - 9$ ✓; the original $T(n - 7, 1) \in \{n - 6, \dots, n\}$ exceeds $n - 7 = T^*(n - 8, 1)$ ✓. Row 1: $T^*(1, 1) = 1 < n - 8 = T^*(1, 2)$ ✓ (since $n - 8 = q - 2 \geq 2$ for $q \geq 4$); $T^*(1, 2) = n - 8 < T(1, 3) \in \{n - 6, \dots, n\}$ ✓. Column 2: $T^*(1, 2) = n - 8 < T(2, 2) \in \{n - 6, \dots, n\}$ ✓. So T^* is a valid SYT of λ .

Prefix drop forces $T^* \notin \text{supp}(B)$. $T^*(n - 8, 1) = n - 7 \neq n - 8$, so $p(T^*) \leq n - 9 < n - 8$. By Theorem 11, every $U \in \text{supp}(B)$ has $p(U) \geq n - 8$, hence $T^* \notin \text{supp}(B) \supseteq \text{supp}(w)$.

Coefficient contradiction. Compute

$$[v_{T^*}](T_{j_0}w) = \sum_{U \in \text{supp}(w)} C_U \cdot [v_{T^*}](T_{j_0}v_U).$$

The diagonal contribution $[v_{T^*}] T_{j_0} v_{T^*}$ requires $T^* \in \text{supp}(w)$ (no). The off-diagonal contribution from v_U requires $s_{j_0} U = T^*$, i.e., $U = s_{j_0} T^* = T$. Hence

$$[v_{T^*}] T_{j_0} w = C_T \cdot \beta \neq 0,$$

while $[v_{T^*}](-w) = -C_{T^*} = 0$. This contradicts $T_{j_0} w = -w$. Hence no $T \in \text{supp}(B)$ has $p(T) = n - 8$ exactly, and every such T has $p(T) \geq n - 7$. $\square$

Theorem 14 (Sharpness, computationally verified). *For $\lambda = (3, 3, 1^q)$ at every $q \in \{4, 5, 6, 7\}$, $\text{supp}(B)$ contains an SYT T with $p(T) = n - 8 = q - 2$ exactly; consequently $B \not\subseteq E_{q-2}^-|_{V_\lambda}$.*

Proof. Existence is verified computationally (`~/projects/scratch/2026-05-14-extended-sign-kill-33/explo`). An explicit witness at $q = 4$ ($n = 10$, $p(T) = 2$):

$$T(1, \cdot) = (1, 3, 4), \quad T(2, \cdot) = (2, 5, 6), \quad T(3, 1) = 7, \quad T(4, 1) = 8, \quad T(5, 1) = 9, \quad T(6, 1) = 10.$$

Column 1 entries are $(1, 2, 7, 8, 9, 10)$, so $p(T) = 2$ exactly.

Combined with Lemma 13, $B \subseteq E_{q-2}^-$ would force every $U \in \text{supp}(B)$ to have $p(U) \geq q - 1$, contradicting the witness T with $p(T) = q - 2$. Hence $B \not\subseteq E_{q-2}^-$. $\square$

Remark 15 (Stronger empirical statement). Computational verification (mod $P = 100\,003$, $Q = 11$) at $q \in \{4, 5, 6, 7\}$ shows that $S_{q-2}|_B$ has full rank $2 = \dim B$ at $j_0 = q - 2$, i.e., $B \cap E_{j_0}^- = 0$ (not merely $B \not\subseteq E_{j_0}^-$). Theorem 14 proves the weaker non-containment.

7 Eigenspace summary

Theorem 16 (Eigenspace summary, $\hat{\lambda} = (3, 3)$). *For $\lambda = (3, 3, 1^q)$ with $q \geq 4$,*

$$B \subseteq E_\ell^+ \cap \bigcap_{j=1}^{q-3} E_j^-|_{V_\lambda}, \quad B \not\subseteq E_{q-2}^-|_{V_\lambda}.$$

The E_ℓ^+ containment is sharp with scalar action: $T_\ell|_B = q \cdot \text{Id}_B$ (May-13 evening Theorem A).

Proof. E_j^- chain at $j \leq q - 3$: Theorem 11. Non-containment at $j = q - 2$: Theorem 14. E_ℓ^+ containment with scalar action: May-13 evening (universal across all λ in stable regime). $\square$

The eigenspace constraints alone do not pin down B : computationally the intersection on the right has dimension strictly $> \dim B = 2$ in every tested case. The dimension formula $\dim B = f^{\lambda^{(\geq 2)}} = f^{(2,2)} = 2$ (May-13 multi-corner) is a sharper fact.

8 Computational verification

Theorem 17 (Verification). *Theorem 16 was verified computationally at $q \in \{2, 3, 4, 5, 6, 7\}$ (i.e., $n \in \{8, 9, 10, 11, 12, 13\}$), mod $P = 100\,003$ at $Q = 11$. In every case:*

- $\dim B = 2$, matching the May-13 multi-corner formula.
- $|\text{supp}(B)|$ takes values $42, 72, 78, 78, 78, 78$ for $q = 2, \dots, 7$ (stabilizes at 78 from $q = 4$).
- Every $T \in \text{supp}(B)$ has $p(T) \geq q - 2 = n - 8$, with the bound achieved at sharpness witnesses for $q \in \{4, 5, 6, 7\}$.

- $T_j b = -b$ for every $b \in B$ and $1 \leq j \leq q - 3$.
- $S_{q-2}|_B$ has rank $\dim B = 2$ (stronger than Theorem 14; cf. Remark 15).
- The two-piece decomposition $E_{n-1}^+ = \Phi_A(V_{\mu_A}) \oplus \Phi_{\text{hook}}(V_{\mu_{\text{hook}}})$ verified directly: $\dim R'_n V_\lambda = f^{\mu_A} + f^{\mu_{\text{hook}}}$ in every case.
- The hook piece's pullback contribution $R'_{n-4}^{(\mu_{\text{hook}})} B_{n-3}^{(\mu_{\text{hook}})} V_{\mu_{\text{hook}}} = 0$ verified at every $q \geq 2$.
- Inner μ_A piece: $\dim B_{n-4}^{(\mu_A)} = 2$, with min prefix $= q - 2$, matching the recursive $(3, 2)$ theorem at $q' = q - 1$.

Proof. Scripts in `~/projects/scratch/2026-05-14-extended-sign-kill-33/`:

- `explore_33_family.py` — main verification ($\dim B$, support prefix, E_j^- chain, recursive $(3, 2)$ input).
- `check_reduction.py` — two-piece decomposition of E_{n-1}^+ and dimension match.
- `check_samerow_contribution.py` — existence of same-row tableaux in $\text{supp}(B)$ (cf. Remark 8).

□

□

9 Discussion

9.1 What is new

This paper closes the $\widehat{\lambda} = (3, 3)$ row of the extended sign-kill conjecture (May-14 evening) in the containment direction at the predicted threshold $\tau = q - 3$, and proves sharpness at $j = q - 2$ (modulo a computational boundary-witness input, Theorem 14, verified for $q \in \{4, 5, 6, 7\}$). Prior $\widehat{\lambda}$ -table entries:

- $(2, 2)$ — May-14 fourth pass.
- $(3, 2)$ — May-14 fifth pass.
- $(4, 2)$ — May-14 sixth pass.

The new entry $(3, 3)$ shares the predicted $\tau = q - 3$ with $(4, 2)$.

9.2 The $r = 1$ vs. $r = 2$ distinction

$\widehat{\lambda} = (3, 3)$ is an $r = 1$ multi-corner shape (only two removable corners), in contrast to the $r = 2$ shapes $(3, 2, 1^q)$, $(4, 2, 1^q)$, $(4, 3, 1^q)$, etc. This changes the reduction topology in two ways:

1. There is only *one* doubly-stripped partition (not three), namely $\mu_A = \lambda \setminus \{c_1, c_2\}$. Equivalently: only one 2-block Phi piece.
2. The same-row $+q$ -eigenvectors at c_1 (when c_1 has a left-adjacent cell in the same row, as here) constitute an additional Phi-piece with hook inner shape, which must be handled separately. This piece is structurally the analogue of the C -piece in $r = 2$ reductions and likewise *vanishes after pullback*.

The end result is the cleanest reduction yet: $B = (T_{n-4} + 1)(T_{n-3} + 1)(T_{n-2} + 1) \Phi_A(B_{n-4}^{(\mu_A)} V_{\mu_A})$ with a single recursive input (the $(3, 2)$ theorem).

9.3 The same-row Phi issue and the $(4, 2)$ gap

The discovery (Remark 8) that the sixth-pass $(4, 2)$ paper's Lemma 3.1(ii) overstates the $+q$ -eigenspace decomposition is a side-product of the present work. The gap is fixable by an additional same-row vanishing argument analogous to ours; in the $(4, 2)$ case the required input is the $(2, 2)$ extended sign-kill theorem (May-14 fourth pass). A companion erratum/extension note will close this.

9.4 Toward the next family

With $\widehat{\lambda} = (3, 3)$ now closed, the table of proved extended sign-kill rows is:

$\widehat{\lambda}$	predicted τ	status
$(2, 2)$	$q - 1$	proved (May-14 fourth pass)
$(3, 2)$	$q - 2$	proved (May-14 fifth pass)
$(3, 3)$	$q - 3$	proved (this paper)
$(4, 2)$	$q - 3$	proved (May-14 sixth pass; same-row gap to close)
$(4, 3)$	$q - 4$	open (fully recursive: needs $(3, 3), (4, 2), (3, 2)$)
$(5, 2)$	$q - 4$	open (recursive on $(4, 2)$ + hook)
$(3, 3, 2)?$	$?$	probably out of stable regime

The next natural target $\widehat{\lambda} = (4, 3)$ is the first *fully recursive* case: all three doubly-stripped pieces are non-hooks, each handled by a recursive extended sign-kill input from the table above.

9.5 What this is not

This paper continues the structural sharpening of B via its T_j -eigenspace containments — a programme orthogonal to (and unaffected by) the iso-programme refutations of May-13 and May-14 (morning/afternoon). We learn more about *where* B sits inside V_{λ} via these eigenspace conditions, without claiming B equals any standard representation-theoretic object.

10 Status

Established (this paper).

- Lemma 7: two-piece decomposition of E_{n-1}^+ (the new structural ingredient for $r = 1$ shapes with a same-row corner).
- Theorem 9: single-branch reduction (after the hook piece vanishes), $\lambda = (3, 3, 1^q)$, $q \geq 2$.
- Theorem 11: $B \subseteq E_j^-$ for $j \leq q - 3$ at $q \geq 4$.
- Theorem 14: sharpness at $j = q - 2$ for $q \in \{4, 5, 6, 7\}$ (with Lemma 13's structural reduction at all $q \geq 4$).
- Theorem 16: complete eigenspace summary.
- Discovery: $(4, 2)$ paper Lemma 3.1(ii) overstates the E_{n-1}^+ decomposition; the gap is closable via the $(2, 2)$ extended sign-kill input (companion note pending).

Open.

- Structural (non-computational) sharpness across all $q \geq 4$ (same residual obstacle as the fifth- and sixth-pass papers: existence of a boundary-prefix witness in $\text{supp}(B)$).
- Sharpness at $j \in \{q-1, q, \dots, n-1\} \setminus \{\ell\}$ (consistent computationally; argument should be analogous to the fifth-pass paper).
- The next family $\widehat{\lambda} = (4, 3)$ at $\lambda = (4, 3, 1^q)$ (fully recursive three-branch, no hook input).
- General $\widehat{\lambda}$ at $\ell(\widehat{\lambda}) \geq 2$: the support prefix bound $p(T) \geq q + 3 - \widehat{\lambda}_1 - \widehat{\lambda}_2$ across $\text{supp}(B)$ (May-14 fifth-pass Conjecture 9).
- $(4, 2)$ companion erratum closing Lemma 3.1(ii) gap.

Push status. Pending PAT refresh; prior unpushed-commit count on `clio-vega/proofs`: 67 (after sixth-pass commit).